

0.1 Section 7.1 - Intro to Probability

0.1.1 Event and Sample Space

Definition 1 (Event and Sample Space)

Let S be a finite, non-empty set called a **sample space** and a set $E \subseteq S$ called the **event space**. The probability of E occurring is defined as

$$P(E) = \frac{|E|}{|S|}$$

Example 1

Let's do some examples

1. Determine the probability of rolling the following numbers when 2 dice are rolled

(a) 12

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 12, $|E| = 1$

$$\begin{aligned} P(E) &= \frac{|E|}{|S|} \\ &= \boxed{\frac{1}{36}} \end{aligned}$$

(b) 2

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 2, $|E| = 1$

$$\begin{aligned} P(E) &= \frac{|E|}{|S|} \\ &= \boxed{\frac{1}{36}} \end{aligned}$$

(c) 3

$$\begin{aligned}
 S &= \text{possible rolls of 2 dice, } |S| = 36 \\
 E &= \text{rolls that give a 3, } |E| = 2, (1 + 2), (2 + 1) \\
 P(E) &= \frac{|E|}{|S|} \\
 &= \frac{2}{36} \\
 &= \boxed{\frac{1}{18}}
 \end{aligned}$$

(d) 4

$$\begin{aligned}
 |S| &= 36 \\
 E &= \{(1, 3), (2, 2), (3, 1)\} \\
 |E| &= 3 \\
 P(E) &= \frac{3}{36} \\
 &= \boxed{\frac{1}{12}}
 \end{aligned}$$

This is just

Sum of dice	Probability
2	$\frac{1}{36}$
3	$\frac{2}{36}$
4	$\frac{3}{36}$
5	$\frac{4}{36}$
6	$\frac{5}{36}$
7	$\frac{6}{36}$
8	$\frac{5}{36}$
9	$\frac{4}{36}$
10	$\frac{3}{36}$
11	$\frac{2}{36}$
12	$\frac{1}{36}$

2. Determine the probability of rolling a 2 when a single die is rolled

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$E = \{2\}$$

$$P(\text{rolling } 2) = \boxed{\frac{1}{6}}$$

3. Determine the probability of not rolling a 2 when a die is rolled

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$E = S - \{2\}$$

$$= \{1, 3, 4, 5, 6\}$$

$$P(\text{not rolling } 2) = \boxed{\frac{5}{6}}$$

or

$$P(\text{not rolling } 2) + P(\text{rolling } 2) = 1$$

$$P(\text{not rolling } 2) = 1 - P(\text{rolling } 2)$$

$$= 1 - \frac{1}{6}$$

$$= \boxed{\frac{5}{6}}$$

4. A card is chosen from a standard 52 deck. Determine the probability that the card is
- (a) An ace

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= 4 \\
 P(\text{ace}) &= \frac{4}{52} \\
 &= \boxed{\frac{1}{13}}
 \end{aligned}$$

(b) An spade

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= 13 \\
 P(\text{spade}) &= \frac{13}{52} \\
 &= \boxed{\frac{1}{4}}
 \end{aligned}$$

(c) A spade or an ace

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= |\text{Ace} \cup \text{Spade}| \\
 &= |\text{Ace}| + |\text{Spade}| - |\text{Ace} \cap \text{Spade}| \\
 &= 4 + 13 - 1 \\
 &= 16
 \end{aligned}$$

$$\begin{aligned}
 P(\text{ace or spade}) &= \frac{16}{52} \\
 &= \boxed{\frac{4}{13}}
 \end{aligned}$$

0.1.2 Probability with Multiple Subsets (OR)

From these examples we see that if $E_1, E_2 \subseteq S$, then

$$\begin{aligned} P(E_1 \cup E_2) &= \frac{|E_1 \cup E_2|}{|S|} \\ &= \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|} \end{aligned}$$